WEEK 5 REPORT

Power BI – Healthcare Data Cleaning and Data Modeling

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Course: B.Sc. Data Science and AI

This report explains the Week 5 work completed in Power BI. The main tasks were cleaning the healthcare dataset, creating useful columns and tables, merging data, creating DAX measures, and building relationships between tables.

The same Power BI screenshots from the original report are included as evidence of the work.

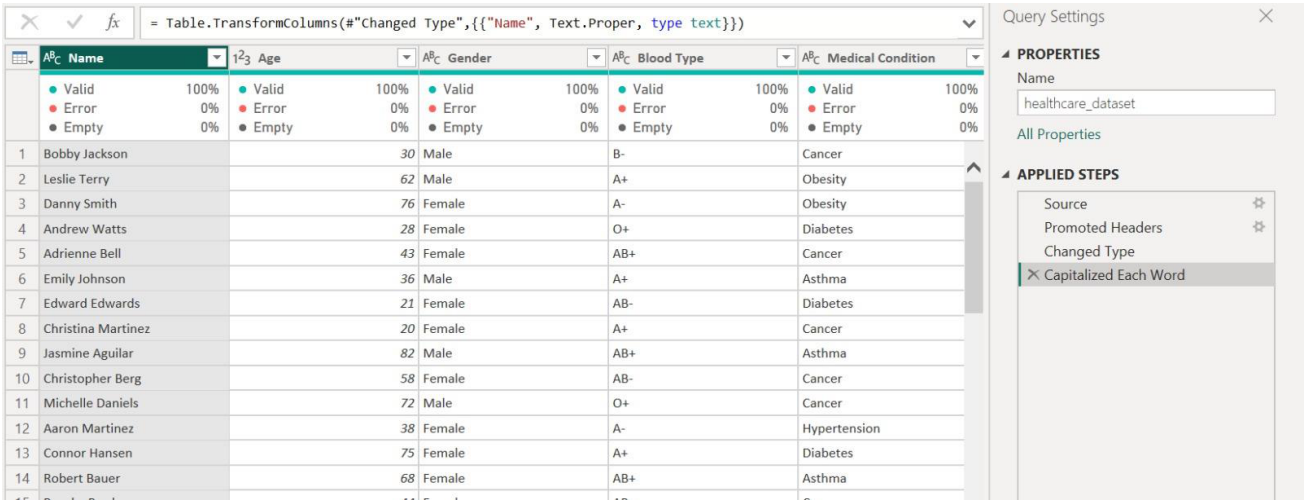
1. Data Cleaning

Capitalize Each Word

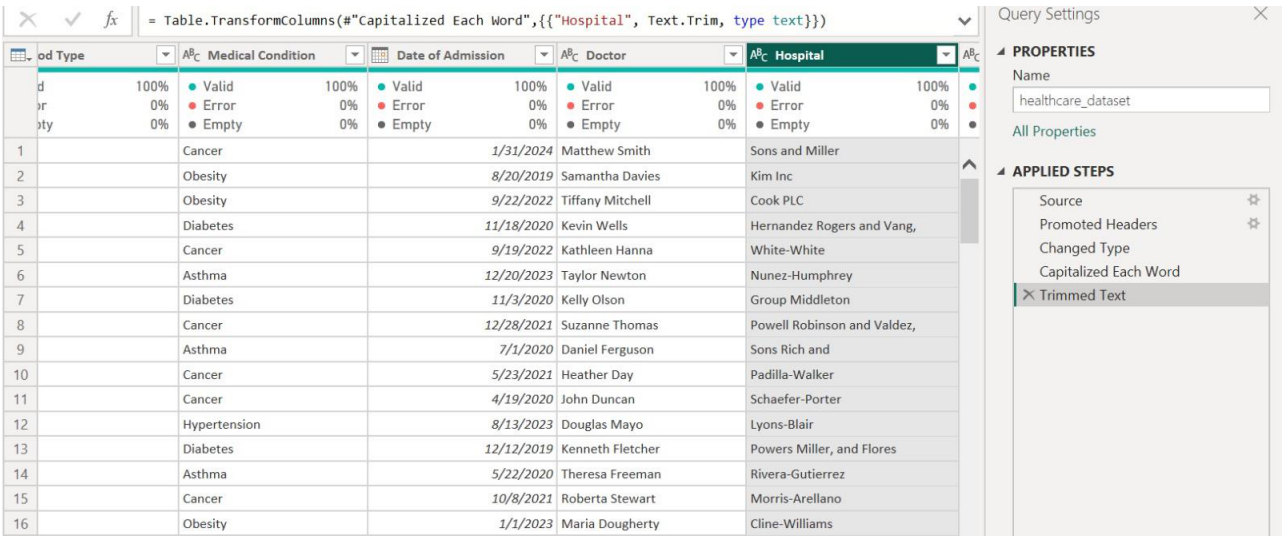
The text values were changed to a proper format using **Capitalize Each Word**. This makes names and other text values easier to read and keeps the data consistent.

Trimmed Text

Extra spaces in the text columns were removed using the **Trim** transformation. This helps avoid unwanted spaces and improves data consistency.



	Name	Age	Gender	Blood Type	Medical Condition
1	Bobby Jackson	30	Male	B-	Cancer
2	Leslie Terry	62	Male	A+	Obesity
3	Danny Smith	76	Female	A-	Obesity
4	Andrew Watts	28	Female	O+	Diabetes
5	Adrienne Bell	43	Female	AB+	Cancer
6	Emily Johnson	36	Male	A+	Asthma
7	Edward Edwards	21	Female	AB-	Diabetes
8	Christina Martinez	20	Female	A+	Cancer
9	Jasmine Aguilar	82	Male	AB+	Asthma
10	Christopher Berg	58	Female	AB-	Cancer
11	Michelle Daniels	72	Male	O+	Cancer
12	Aaron Martinez	38	Female	A-	Hypertension
13	Connor Hansen	75	Female	A+	Diabetes
14	Robert Bauer	68	Female	AB+	Asthma

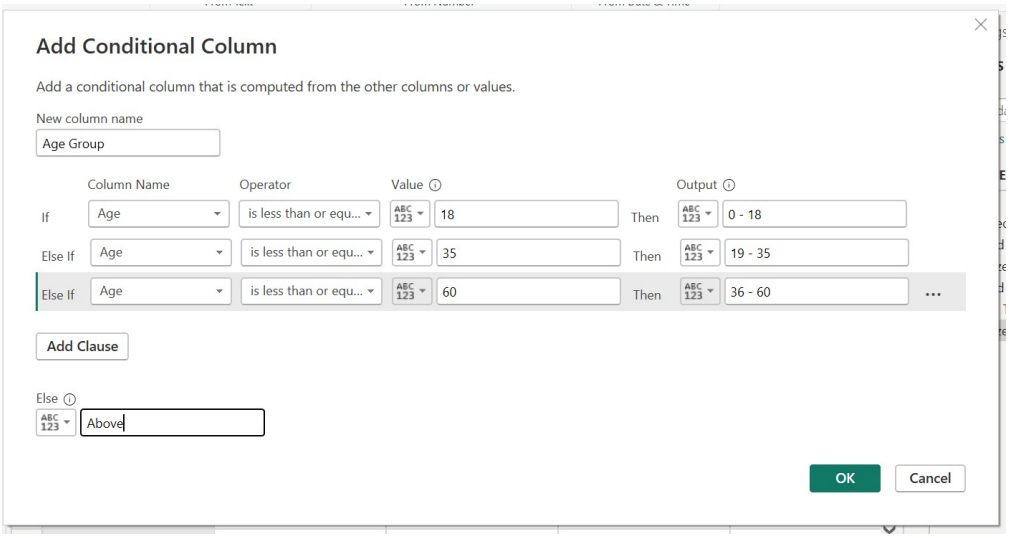


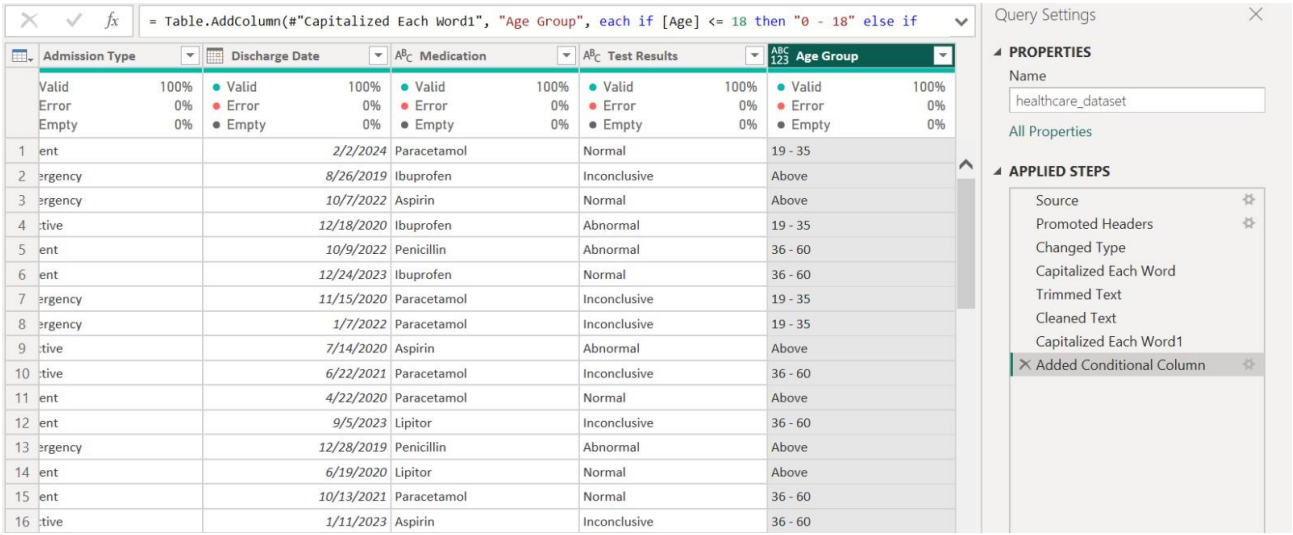
	od Type	Medical Condition	Date of Admission	Doctor	Hospital
1		Cancer	1/31/2024	Matthew Smith	Sons and Miller
2		Obesity	8/20/2019	Samantha Davies	Kim Inc
3		Obesity	9/22/2022	Tiffany Mitchell	Cook PLC
4		Diabetes	11/18/2020	Kevin Wells	Hernandez Rogers and Vang,
5		Cancer	9/19/2022	Kathleen Hanna	White-White
6		Asthma	12/20/2023	Taylor Newton	Nunez-Humphrey
7		Diabetes	11/3/2020	Kelly Olson	Group Middleton
8		Cancer	12/28/2021	Suzanne Thomas	Powell Robinson and Valdez,
9		Asthma	7/1/2020	Daniel Ferguson	Sons Rich and
10		Cancer	5/23/2021	Heather Day	Padilla-Walker
11		Cancer	4/19/2020	John Duncan	Schaefer-Porter
12		Hypertension	8/13/2023	Douglas Mayo	Lyons-Blair
13		Diabetes	12/12/2019	Kenneth Fletcher	Powers Miller, and Flores
14		Asthma	5/22/2020	Theresa Freeman	Rivera-Gutierrez
15		Cancer	10/8/2021	Roberta Stewart	Morris-Arellano
16		Obesity	1/1/2023	Maria Dougherty	Cline-Williams

2. Age Group

Creating Age Group

An **Age Group** column was created using a Conditional Column. Patients were grouped based on their age so that the data could be analysed more easily.





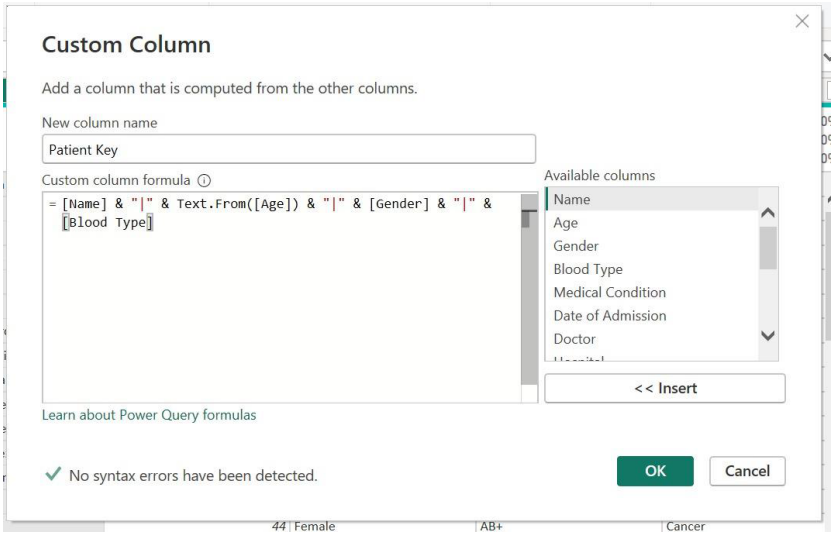
3. Patient Key

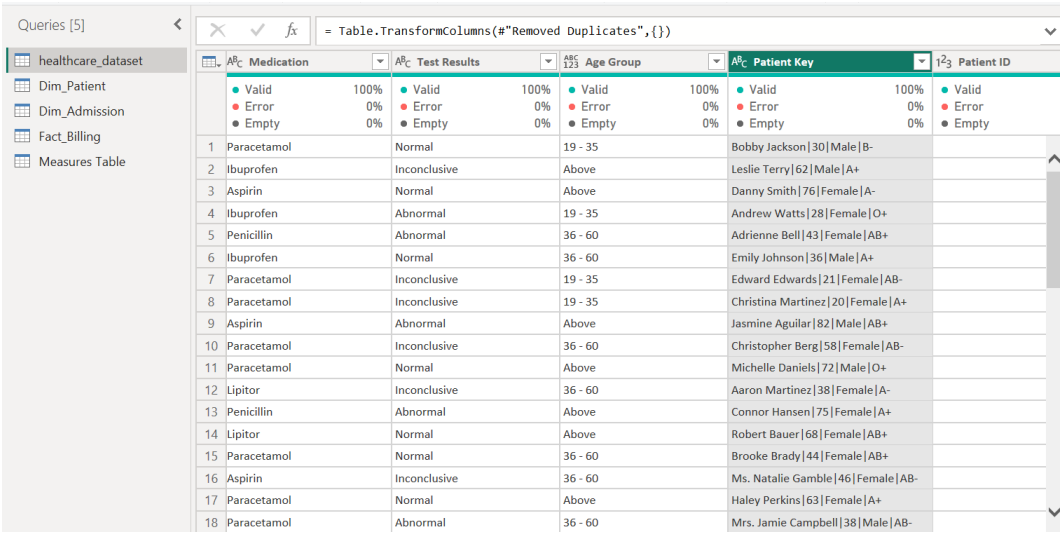
Creating Patient Key

A custom column called **Patient Key** was created to identify patients using a combination of important details.

Formula

```
[Name] & "|" & Text.From([Age]) & "|" & [Gender] & "|" & [Blood Type]
```





	Medication	Test Results	Age Group	Patient Key	Patient ID
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	Paracetamol	Normal	19 - 35	Bobby Jackson|30|Male|B-	
2	Ibuprofen	Inconclusive	Above	Leslie Terry|62|Male|A+	
3	Aspirin	Normal	Above	Danny Smith|76|Female|A-	
4	Ibuprofen	Abnormal	19 - 35	Andrew Watts|28|Female|O+	
5	Penicillin	Abnormal	36 - 60	Adrienne Bell|43|Female|AB+	
6	Ibuprofen	Normal	36 - 60	Emily Johnson|36|Male|A+	
7	Paracetamol	Inconclusive	19 - 35	Edward Edwards|21|Female|AB-	
8	Paracetamol	Inconclusive	19 - 35	Christina Martinez|20|Female|A+	
9	Aspirin	Abnormal	Above	Jasmine Aguilar|82|Male|AB+	
10	Paracetamol	Inconclusive	36 - 60	Christopher Berg|58|Female|AB-	
11	Paracetamol	Normal	Above	Michelle Daniels|72|Male|O+	
12	Lipitor	Inconclusive	36 - 60	Aaron Martinez|38|Female|A-	
13	Penicillin	Abnormal	Above	Connor Hansen|75|Female|A+	
14	Lipitor	Normal	Above	Robert Bauer|68|Female|AB+	
15	Paracetamol	Normal	36 - 60	Brooke Brady|44|Female|AB+	
16	Aspirin	Inconclusive	36 - 60	Ms. Natalie Gamble|46|Female|AB-	
17	Paracetamol	Normal	Above	Haley Perkins|63|Female|A+	
18	Paracetamol	Abnormal	36 - 60	Mrs. Jamie Campbell|38|Male|AB-	

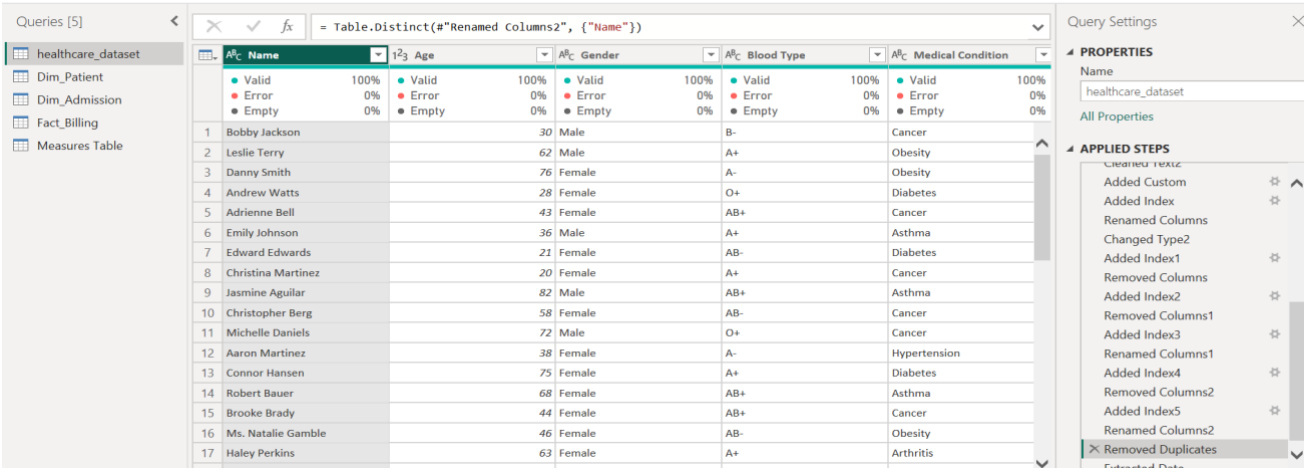
4. Dim_Patient Table

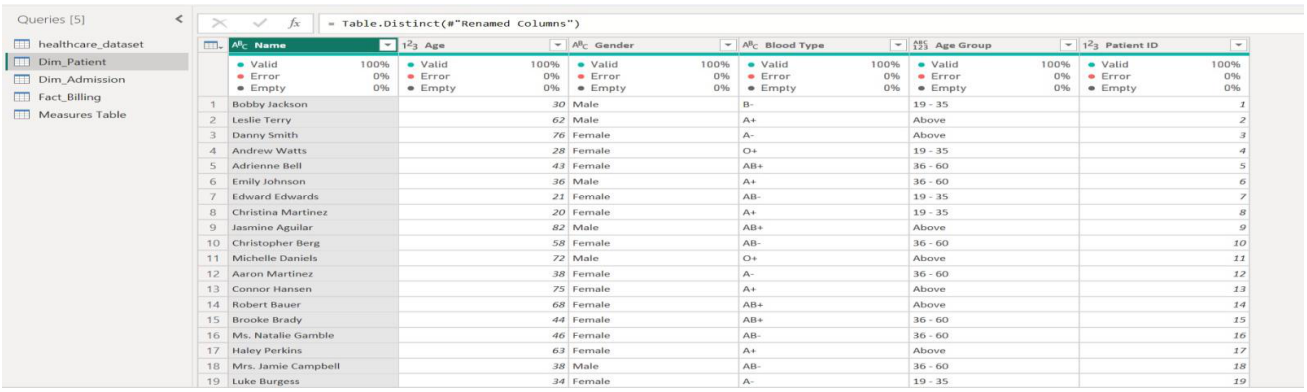
Removing Duplicates

Duplicate patient records were removed to create a clean patient table.

Dim_Patient

The cleaned table was named **Dim_Patient**. It contains unique patient information that can be used in the data model.





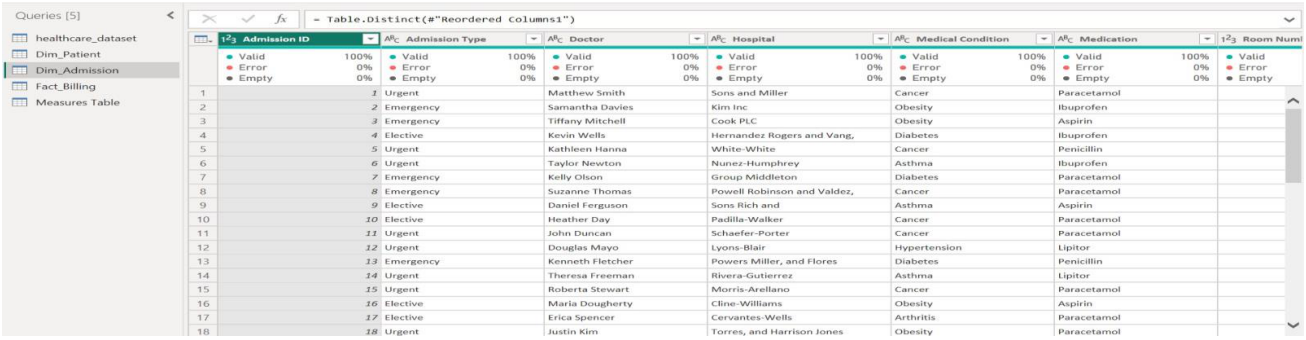
5. Dim_Admission and Fact_Billing

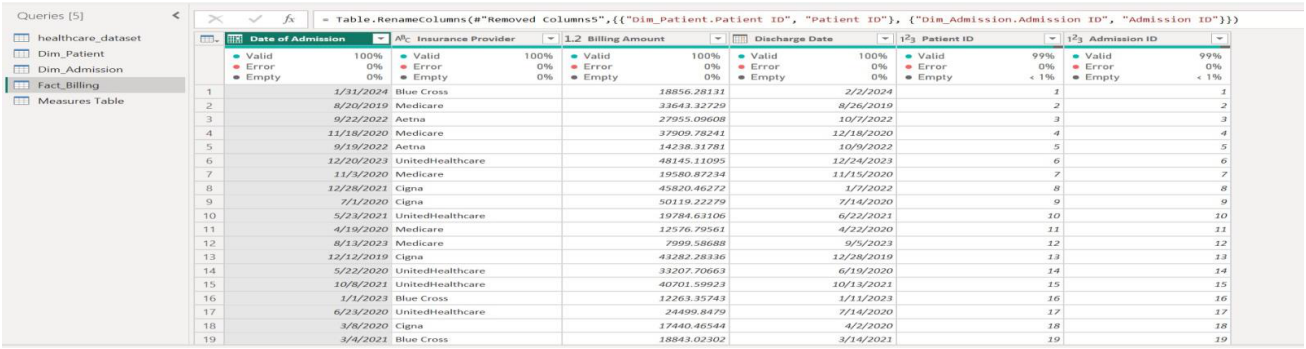
Dim_Admission

A separate **Dim_Admission** table was created to store admission-related information.

Fact_Billing

A **Fact_Billing** table was created to store billing and other measurable information. These tables help organize the healthcare data for analysis.

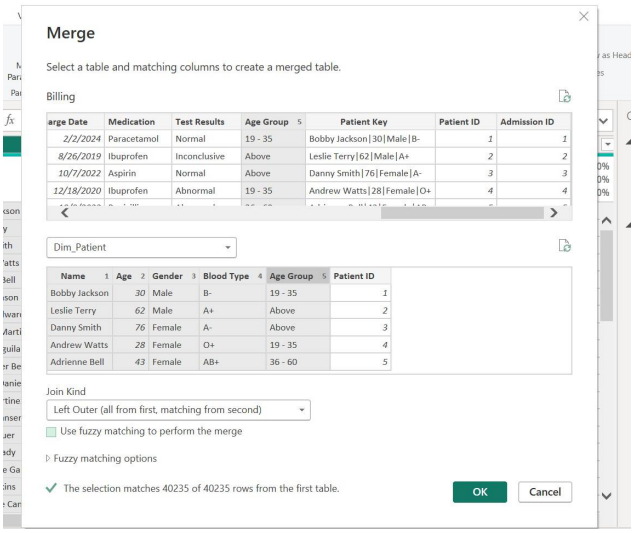




6. Merge with Dim_Patient

Merging Tables

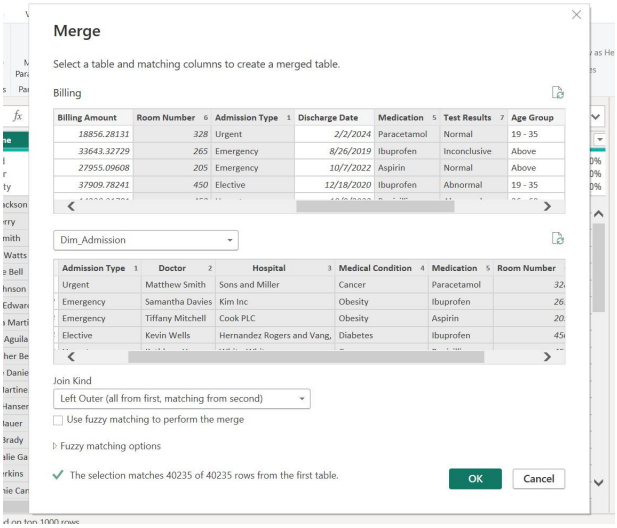
The main healthcare dataset was merged with **Dim_Patient** using the Patient ID. The merge showed **40,235 of 40,235** records matched.



7. Merge with Dim_Admission

Merging Tables

The healthcare dataset was also merged with **Dim_Admission** using the Admission ID. The merge showed **40,235 of 40,235** records matched.



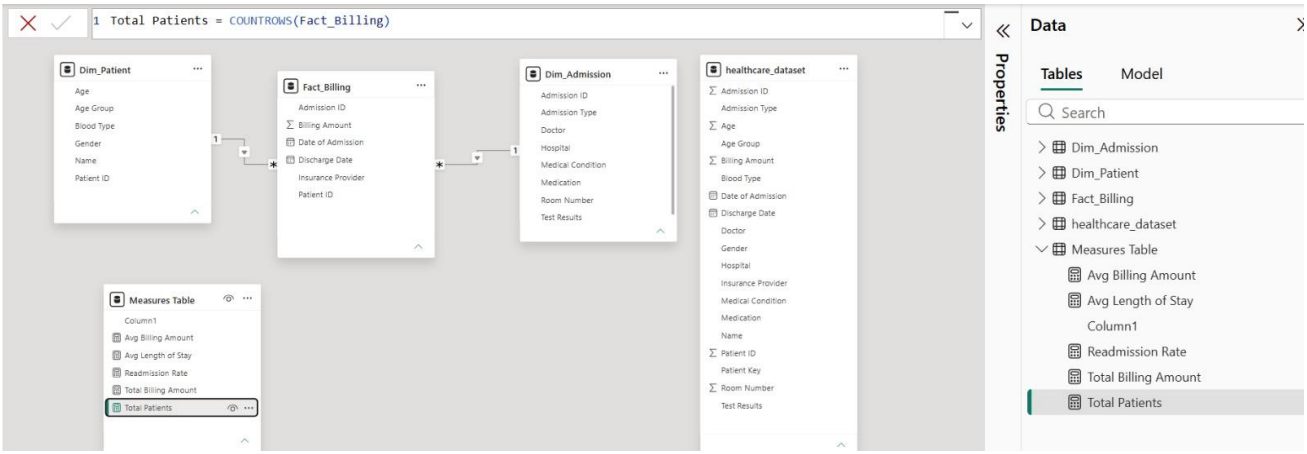
8. DAX Measures

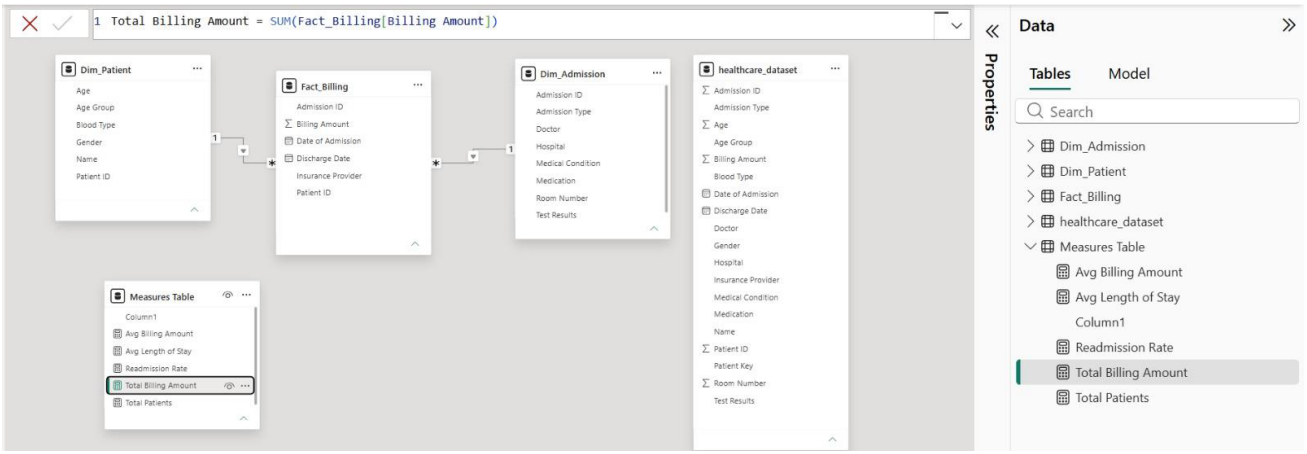
Total Patients

A DAX measure was created to calculate the total number of patients.

Total Billing Amount

Another DAX measure was created to calculate the total billing amount.





9. Average Measures

Average Billing Amount

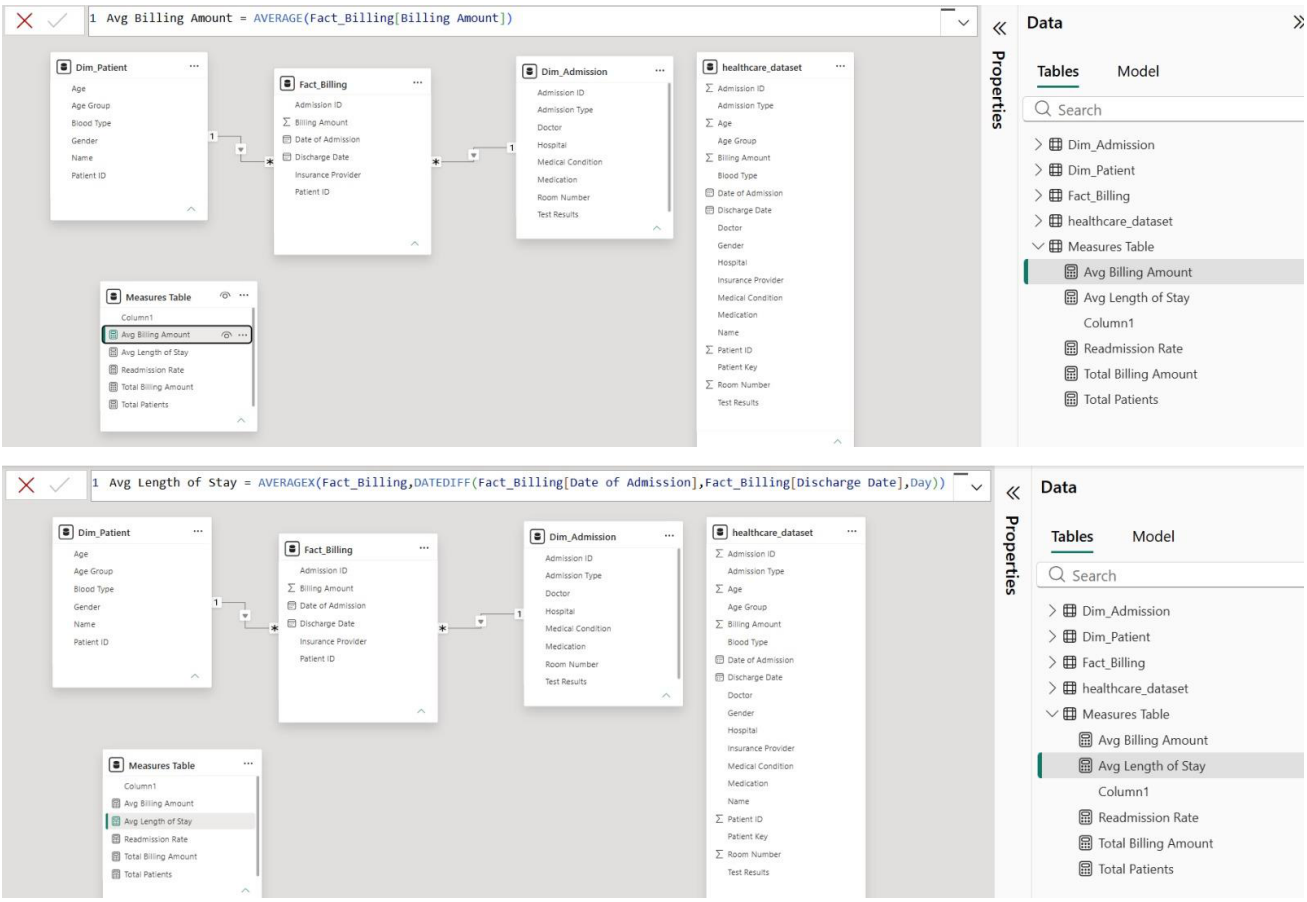
A DAX measure was created to calculate the average billing amount.

Average Length of Stay

A DAX measure was created to calculate the average number of days patients stayed in the hospital.

Readmission Rate

A DAX measure was created to calculate the percentage of patients who were readmitted.



10. Relationships and Conclusion

Creating Relationships

Relationships were created between the tables using common fields such as Patient ID and Admission ID. This connects the tables and allows the data to work together in Power BI.

Conclusion

In Week 5, the healthcare dataset was cleaned and organized in Power BI. Important columns, dimension tables, and fact tables were created. The tables were merged and connected using relationships. DAX measures were also created for useful calculations such as patient count, billing amount, average billing, length of stay, and readmission rate.

